eiRxiv EMERGING INVESTIGATORS PREPRINT SERVER

Instructions for Authors (first two pages):

We have created this manuscript template to aid our student authors in the formatting of their work when submitting to *eiRxiv*.

Using This Template:

- The following pages contain a manuscript template for you to use for your submission.
    - **Please do not simply copy and paste your text onto this template**.
        - Take time to read the examples and advice given for each section.
    - **Do NOT change the order of the sections, add any sections, or remove any sections**. You must use the sections and order as laid out on the following pages.

- We have provided information under each heading to guide authors as well as point out common mistakes seen.
    - Our online submission guide contains more information than presented here. We strongly encourage all authors to read this guide throughout the submission process. *It will save you time during the submission process.*

- As you write, remove any non-heading text (*in italics)* in this template.
    - All that should be left in the end are the headings and your text (non-italic, unless needed) telling us about your research!

General Formatting Items:

- 1.5x Line Spacing
- Font: Arial, size 11
- 1-inch margins on all sides
- 10 page limit
    - o Counted from the start of the Introduction to the end of the Materials/Methods
    - o At initial submission you may be 1-1.5 pages over this limit as we work with you on manuscript content and refining the material presented.
- 8 Table + Figure maximum
- Please leave the line numbers on the document – it helps out our reviewers and editors!
- Do not use specially formatted headings or subheadings – they should all be bolded 11pt font

Before Submitting:

- Ensure that all authors have given permission for this work to be published at *EIRXIV.*

- All images and tables are original to this work
    - **Any image and/or table published in other sources required approval for copyright reasons**. Please reach out to us prior to submission if you are considering using a previously published image/table.

- You have all study approval documents (if applicable)
    - Please see our Human and Animal Subjects guidelines for more information
    - For consent forms, please upload just the blank version.
        - If we need to see filled out forms, we will contact you after submission.

- Lastly, please delete the “Instructions for Authors” pages from the manuscript file

Using eiRxiv Submission Site:

- The senior/corresponding author (or parent/guardian) must create an account.
    - o If a parent/guardian is submitting the manuscript and is not the senior author, please ensure you have received permission from the senior author to publish.
    - o The individual who submits the work will be who all emails from us are addressed to. This does NOT reflect the actual order of authors which will be pulled from your manuscript Word file.

- The student author is allowed to submit their manuscript to *eiRXIV* using their own eiRxiv submission account.
- An adult's approval is required for any manuscript submission.

- The manuscript file needs to be in PDF document format

- Figures must be saved as jpg/png/tiff files in order to be recognized by the system
    - Figures should be uploaded separately in Editorial Manager **as well as placed at the end of this template above their captions**
    - If a figure has multiple panels (i.e., Figure 1a, b, c), it must be uploaded it as **one** file.

- Any study approval documents should be uploaded under "other"

- Check out our full submission checklists for all other items needed to submit

Contacting the Staff:
Should you have any questions throughout the submission process, please contact us at submissions@eiRxiv.org. If you have submitted, please include the title, and, if accepted for review, the manuscript number (Ex: EIRXIV-22-153) in any email to the staff.

Single Cell RNA-Seq Analysis of microglial activation: region-specific immune activation in Alzheimer's Disease

First Author[1], Second Author[2], Third Author[1,2]

Abhirup Mukherjee, Folsom High School, Folsom, California,

[1] Department (if applicable), Institution/School, City, State, Country (if non-US)
[2] Department (if applicable), Institution/School, City, State, Country (if non-US)
[3] Department (if applicable), Institution/School, City, State, Country (if non-US)

Affiliations of "unaffiliated", "mentor", or that only give general industry/position (i.e., tech sector) are NOT allowed. Authors may default to the student author's affiliation if needed.

Student Authors
Please list all student author names here and indicate whether in middle or high school

Authorship Guidelines:
*Equal contributions should be noted by placing an * after each author who contributed equally. The statement "*These authors contributed equally to this work" should be given at the end of the author affiliations.*

A senior (adult) author is required for all EIRXIV manuscripts. This person is generally the individual who supervised the research. They often don't participate in the data collection but provide feedback on experimental design and in writing the manuscript.

Please do not list degrees for any of the authors on the title page as it will be removed before publishing.

SUMMARY

Alzheimer’s and Parkinson’s Diseases are some of the most prevalent neurodegenerative disorders worldwide. The loss of neuronal function inflicts detrimental damage, especially to those of age, and it is imminent that a cure be found. Cell types, such as neurons and microglia, play a key role in disease progression. While many regions of the brain, such as the hippocampus have been investigated to study cell-type specific functions, it is pivotal that other regions, such as the Middle Temporal Gyrus, are also investigated to find a solution. Different RNA-Seq analytical methods, such as gene set enrichment analysis, quality control, and dimensionality reduction resulted in findings that prove that microglia largely maintained their structural integrity while exhibiting little transcriptional variability. The aim of this study was to identify region and cell-specific transcriptional patterns in the event of neurodegeneration. To further investigate the transcriptional variability, gene set enrichment analysis and differential expression were used to identify sets of upregulated and downregulated processes and genes involved for both Alzheimer's and Parkinson’s disease samples. This revealed cell-specific activations under neural stress. Specifically, neurons showed regulations in synaptic and metabolic processes in Alzheimer’s and Parkinson’s samples, while microglia showed regulation in their immune pathways. Additionally, a single cell RNA-sequencing analysis was made from data for over 40,000 microglial cells in the Middle Temporal Gyrus. Using the previous methods such as Quality Control, dimensionality reduction, and more, the findings helped us underscore region and cell specific transcriptome profiling which may prove beneficial in targeting neural dysfunctionality.

INTRODUCTION

Alzheimer’s Disease and Parkinson’s Disease are some of the most infamous neurodegenerative disorders in the world, because of their prevalence. They are both disorders characterized by cognitive and motor decline, respectively, affecting millions of people worldwide(American Brain Foundation). The decline is caused by constant decline of synaptic connectivity and neuronal integrity.

Alzheimer’s Disease is commonly characterized by memory impairment, constituting from buildup of proteins clustering in formations known as amyloid plaques(Mayo Clinic). These amyloid plaques that are built up disrupt cell to cell communication, causing neurodegeneration(Mayo Clinic). Oftentimes, people are affected with age, a common characteristic of dementia. Around 6.9 million people 65 years and older are affected with Alzheimer's, and over 70% of those aged 70 and higher(Mayo Clinic).

Parkinson's Disease, in contrast, is characterized by the degeneration of different cell types in different regions of the brain, namely neurons in different regions. Specifically, the neurons that produce dopamine eventually die, and consequently, the ability to move begins to worsen(National Institute of Neurological Disorders and Stroke). These neurons are notably found in the region called the substantia nigra. The loss of dopamine levels leads to visible symptoms associated with Parkinson's Disease, including tremor, rigidity, bradykinesia, postural instability, and more(National Institute of Neurological Disorders and Stroke).

Microglia are an immune macrophagic cell type housed in the Central Nervous System, known for their crucial role in facilitating the environment and responding to infection(Hammond et al.). In neurodegeneration, specifically Alzheimer's Disease, microglia produce cytokines, further developing the neurodegeneration(Hansen et al., 2018). However, microglia react differently in different regions of the brain in the event of loss of neuronal functionality. For example, different areas of the hippocampus reflected different inflammatory responses produced by the microglia in Alzheimer's Disease(Bachstetter et al.).

However, many studies tend to focus on the hippocampus, and other populated regions, but it is pivotal to also focus on different, unexplored regions of the brain as well, such as the Middle Temporal Gyrus(MTG). The MTG specifically is known for semantic memory, and the visual processing of such(Science Direct). The Middle Temporal Gyrus is especially interesting to research, as it is one of the first regions that exhibit the amyloid deposits that are accustomed with diseases such as Alzheimer's(Pfeil et al.).

In this study, our team analyzed single-cell RNA sequencing (scRNA-seq) data publicly available, from the CellxGene, SEA-AD(Seattle-Alzheimer's Disease) repositories, in order to investigate the role of microglia in Middle Temporal Gyrus in Alzheimer's Disease, and cast comparisons with other cell types and diseases, applying different methodologies associated with scRNA sequencing.

Overall, our findings showed that microglia show region-specific immune activation in the event of neural degeneration. In the middle temporal gyrus specifically, transcriptional and structural integrity is maintained in Alzheimer's Disease. These results suggest that microglial responses are region-specific, highlighting the importance of studying the understudied portions of the brain to further our knowledge in disease mechanisms.

RESULTS

Gene Set Enrichment Findings

In order to find biological pathways that highlighted transcriptional changes for microglia and neurons in Alzheimer's and Parkinson's Diseases, gene set enrichment analysis was performed using Gene Ontology Biological Processes. An FDR cutoff of 0.05 was used for each expressed gene.

The upregulated genes for the microglia displayed a significant enrichment in immune pathways **(Figure 1A)**. Some of the significant immune pathways mentioned were the leukocyte mediated immunity, as well as the innate immune response. These enriched pathways after gene set enrichment analysis were performed indicate the immune system activation in the presence of neurodegenerative disease, which is consistent with brain degeneration.

The downregulated genes for the microglia displayed a significant enrichment in immune response as well as cellular differentiation, resulting in the specialization of some cells (**Figure 1B)**. Specifically, this included myeloid leukocyte and cytokine regulation. Since these genes are downregulated, specific leukocyte and cytokine signaling is heavily impaired, which leads to further neurodegeneration.

Upregulated neurons in both Alzheimer's and Parkinson's diseases were prevalent in synapse and vesicle-related pathways **(Figure 1C)**. Specifically, vesicle docking and synaptic plasticity were extremely enriched among upregulated neurons in Alzheimer's and Parkinson's diseases, which suggest that in the early stages of neurodegenerative disorders, neurons make an effort to preserve communication under distress.

Downregulated neurons in both neurodegenerative disorders showed enrichment in exocytosis and transport processes **(Figure 1D)**. With pathways such as neuronal action potential propagation being enriched in downregulated neurons, it can be assumed that there is a decrease or impaired among neurotransmitters being passed through, which accounts for the miscommunications.

Differential Expression Findings

To sort out and rectify the genes that were significantly expressed, two volcano plots were made by performing a differential expression analysis for both microglia and neurons in Alzheimer's and Parkinson's. An effect size of 1, and an adjusted p-value of 0.05 to determine gene-level significance.

The neuronal volcano displays a very narrow distribution of the gene expression of neurons in the brain in Alzheimer's and Parkinson's Diseases. There were only a few genes that were above the effect size threshold. This included genes such as MALAT1, IL1RAPL1, and DANT2**(Figure 2A)**. These genes are important for RNA regulation, as well as synaptic scaffolding, which ensures the synaptic signals are responsive in neurons. Some of the genes that were downregulated according to the volcano plot included SNAP25 and RASGRF2**(Figure 2A)**. These are gene families responsible for transport of molecules across, as well as transcriptional regulation. As a whole, the volcano plot indicated little alteration in genes.

The microglial dataset on the other hand showed a much broader distribution, with much more genes being significantly upregulated or downregulated in Alzheimer's and Parkinson's. Downregulated genes in the microglia included DPYD, UBE2D3, TAB2**(Figure 2B)**. These genes are responsible for DNA deconstruction, protein labeling, and signaling inflammation genes. Upregulated genes in the microglia include SRGAP25, which are vital for building and regulating connections among cells, especially in neurons**(Figure 2B).**

Quality Control Metrics

In order to make sure that no “wasteful” biological data was considered, quality control was performed to filter the cells that were degenerating, as well as other irrelevant data. All of the cells were evaluated using n_genes_by_counts, total_counts, and Fraction mitochondrial UMI. A 5% mitochondrial threshold was implemented to filter out cells with high mitochondrial content.

The quality control that was performed indicated that the microglial data from the MTG exhibited strong integrity, with little to none degradation. The number of genes found per cell averaged at 2,706, with a minimum of 998, and a maximum of 11,566 genes per cell **(Figure 3A)**. This distribution showed some degree of transcriptional activity among the Microglia, which is proven with a standard deviation of 1020, asserting that there was indeed substantial variability.

The total UMI’s per cell were on average 3538, with a minimum of 2008 UMI’s, and maximum of 5,415 (**Figure 3B**). With a much smaller range, supported with a standard deviation of 518 UMI’s, it is fairly safe to assume that there was reliable gene detection throughout the data set.

The fraction of UMI from the mitochondria was also very low, at 0.4%**(Figure 3C)**, especially considering 5% was made the cutoff for excluding low quality cells. The low mitochondrial percentages indicate that there is integrity among the UMI’s for the Microglia in the Middle Temporal Gyrus, even in the case of Alzheimer’s Disease.

Microglial Subtypes

To characterize potential subtypes of microglia and transcriptional heterogeneity in the middle temporal gyrus, clustering and dimensionality reduction was performed on the microglial dataset. The Leiden-based clustering showcased the subtypes of microglia while the UMAP plot visualized any transcriptional similarities.

The UMAP visualization of the downloaded data from the SEA-AD dataset of microglia in the MTG confirmed that the only cell type studied was the microglia(**Figure 4A**). No outlying populations outside of this particular cell type were shown, confirming that only microglia were examined throughout.

The Leiden clustering method revealed 10 different subgroups of Microglia found in the MTG(**Figure 4B**). Each subtype occupied a different region inside the middle temporal gyrus. Each boundary represents a tight and distinct border between each subtype, revealing a stable cluster formation representing the biological heterogeneity among all the microglia within the sample.

However, the distributions of donor IDs across the UMAP revealed extensive overlap among different cells of different individuals(**Figure 4C**). Such an intermixing represents similar transcriptional files, with no such boundaries or segregation. Such a uniform mix can represent a stable clustering pattern, showing microglial characteristics across all the samples in the cohort.

Disease Versus Control Transcriptome

To compare the transcriptional states between the donors with Alzheimer's and the control, the donor ID's used earlier were incorporated into group cells depending on disease status. The microglial cells were batch corrected and integrated, and a UMAP was made to show any potential overlap and glaring differences.

Microglial cells derived from control donors exhibited a uniform clustering pattern in the Middle Temporal Gyrus across the UMAP visualization(**Figure 5A**). No clustered formation seemed to be less densely populated than others, indicating a stable level of transcription. Additionally no fragmented divisions were revealed, suggesting a consistent microglial state among control individuals.

However, individuals with cells from the Alzheimer's study revealed slightly more fragmented clustering patterns, and minor expansions in density in certain regions(**Figure 5B**). While the overall structure was preserved from the control study, some variations in density levels in different formations suggest a slight transcriptional difference in response to neurodegeneration.

When control, affected, and donors with missing information were all visualized through UMAP, substantial overlap was revealed(**Figure 5C**). There were no pronounced boundaries among the different visualizations, including the samples with missing information, which were integrated into the overlap, not causing any fragmentation. This could potentially reveal similar transcriptional profiles with those of the control, as well as with Alzheimer's Disease.

Gene Set Pathway Discussion

The enrichment patterns for Microglia exemplifies their role in the process of neuroinflammation in neurodegenerative disorders. Some immune processes, such as the macrophage differentiation, can prove that the inflammatory responses are the cause of some neuronal injuries. Activated microglia are often characterized by transcriptional reprogramming, causing those inflammatory phenotypes(Prater et al.). The enrichment of pathways such as macrophage activation microglia upholds an active state in the MTG. This supports previous claims that hypothesize that extended microglial activation is a result of disease growth.

However, the downregulation of microglial genes, causing pathways such as leukocyte differentiation, show an impaired ability of cell types such as microglia to act as a regulator in the presence of neurodegeneration. Such a pattern has been observed in other data sets as well, such as other single-cell RNA-seq studies, which showed the varied data upon microglia and their number of distinct transcriptional profiles, leading to heterogeneity(Sun et al.).

The enrichment pattern for upregulated neurons however, reveals pathways such as synaptic plasticity as well as vesicle docking. Such pathways prove the initial response of upregulated genes of neurons to maintain neurotransmission despite stress due to neurodegeneration(Lovinger). The enrichment of downregulated genes in synaptic transformation pathways, and exocytosis and propagation pathways show a progressive collapse in neuronal function once a certain threshold of function loss is met. Ultimately these pathways suggest that despite initial efforts to prevent progression of degeneration, ultimately cell types are unable to respond appropriately.

Collectively, the findings emphasize the synaptic stress responses of different cell types of neurodegeneration, and also highlight distinct functional priorities among the cell types. The pathway analysis exemplifies the importance of analyzing and comparing specific pathways when investigating disease mechanisms.

Differential Gene Expression Discussion

Under neurodegeneration, there are obvious differences in the differential expressions of microglial and neuronal cell types in the brain. The microglia experienced a broader pattern of differences in comparison to the neurons. This aligns with the microglia being one of the first responders in the events of pathological distress. Some of the downregulated microglial genes visualized in the volcano plot, including DYPD, TAB2, and UBE2D3, are genes responsible for inflammatory signaling, regulation of transcription, and finding signs of protein degradation. Previous studies have showcased that in the event of microglial activation is characterized by immune related genes, leading to higher transcriptional states in neural stress(Hammond et al.). Because of the decreased microglial expression, it can be detrimental to the ability of the microglial to coordinate the necessary inflammatory response in the presence of Alzheimer’s or Parkinson’s disease.

In comparison to microglia, the graph of neurons was much more restricted, with little variation at most. The upregulated genes seen in the graph of neurons, such as MALAT1 and IL1RAPL1 are usually involved in synaptic organization, which can maybe bring discussion about their adjustments made under cellular stress. The downregulated genes involved, such as SNAP25 and RASGRF2, are involved in transportation and neurotransmission. There are previous studies which have exemplified these patterns, showing degradation of transportation and synaptic machinery in functional loss(Gjoneska et al.).

Across multiple regions of the brain, the findings show the transcriptional ranges and differences between both Microglia and Neurons. Specifically, neurons focus more on transportation and synaptic-related genes. Microglia on the other hand focus more on their immune pathways. One specific finding that was interesting was that the SRGAP2 gene was upregulated in microglia. This gene is more commonly known for its expression in neurons, and so this could be an indicator of a potential transcriptional shift in the event of neuronal loss.

As evidence shows how some cell types can undergo a much broader change than others, it is important to identify the cell-type transcriptional shifts that show how different cell type populations react differently to disease growth and progression.

Microglial Integrity Discussion

The findings prove that Microglia in the Middle Temporal Gyrus exhibit little degradation in the event of Alzheimer's Disease, maintaining exceptional transcriptional integrity. This is supported by the minimal Mitochondrial levels from the UMI's. When a cell is dying or degraded, the nuclear RNA begins to leak out, leading to a larger Mitochondrial RNA percentage among the cells(Kim et al.). Low quality cells are determined when the Mitochondrial RNA fraction of a cell meets the 5% threshold(Ilicic et al.). However, the data indicates the microglia in the MTG are healthy.

Such patterns prove a great contrast in other regions in the brain under neurodegenerative circumstances such as Alzheimer's. Microglia in regions such as the Hippocampus are often under great stress, and will result in much higher mitochondrial levels. Such a discrepancy may prove a difference in the integrity of Microglial cells in specific regions in the brain, such as the MTG, as opposed to areas such as the hippocampus(Agrawal and Jha)
The data provided ultimately validates the low mitochondrial levels in the microglia, which can now guide us onto our subsequent steps. Some of the limitations in this aspect of the study were not being able to compare the data with microglial cells in the MTG with other cell types in the Middle Temporal Gyrus, which would have further analyzed the integrity of Microglial cells.

Findings like these are significant as they show that even during neural degeneration, different regions of the brain react differently depending on the cell-type being analyzed, which can further contribute to our understanding of potential disease preventions.

Heterogeneity and Subtype Discussion

The UMAP reveals 10 transcriptionally different microglial subclusters in the MTG. Such clusters and subgroups can emphasize the heterogeneity of different regions within the MTG. The event of such different subclusters in the same area of the brain but different regions is exemplified in other cortical regions, where multiple phenotypes across subtypes of microglia exist(Stratoulias et al.).

The existence of such subclusters can represent specialized states of the cell type, such as a phagocytosis state in the event of cerebral deterioration. The tight boundaries visualized by the UMAP plots can prove evidence of different functions among each subtype, while the extensive overlap of all the samples from the donors suggest that such specializations are not specific to each donor.
Such variation among each donor suggests a specialized state of different subtypes of each microglia in the MTG are in fact biologically intrinsic, rather than being a result of external factors such as infection and disease. This behavior of cells contrasts with other regions, where in the hippocampus, different astrocytes and other microglia formulate different clusters, depending on the disease(Salta et al.). As a result, microglia in the MTG may exhibit a much more stable behavior, even in the event of neural deterioration, preserving their specialized functional abilities.

The heterogeneity seen by the microglial subtypes shows evidence of specialized functional states of different cell types depending on the region of the brain, which may be vital for supporting homeostasis under neurodegeneration.

Alzheimer's Disease Transcriptome Discussion

With such a similarity in the visualizations of the control versus those in the sample affected by Alzheimer’s Disease, it can be ascertained that microglia in the MTG maintain their transcriptional integrity in the event of degeneration or not. Such a confirmation agrees with studies conducted by other researchers, who affirmed that microglia exhibited much more perturbation and change in subcortical regions than cortical regions, such as the MTG, by analyzing cortical and subcortical tissue(Olah et al.).

However, it is important to analyze the subtle differences that lay among both the models. The model for the samples of those with Alzheimer’s indicates that some of the regions had greater densities of some subtypes, revealing a partial activation of immune pathways identified earlier. However, the overall resilience displayed by the microglial subtypes indicate that they do not go through complete transcriptional reprogramming. However, in comparison to prior reports, other cell types in the MTG often experience different responses as a result of disease detection, such as oligodendrocytes, which show a remyelination response(Gabitto et al.).

Ultimately, all the findings as listed above indicate that microglia in the MTG often do not experience as much perturbation as other cell types in other regions in the event of Alzheimer’s Disease. It is important when investigating mental degeneration, and understanding what it is that makes cells so resilient, and the regions in which they show such resilience in. Future analyses can help further prove the mitigation of communications of different cell types, and even understand the genetic differences of different subtypes of the Microglia, as well as the purposes of the different functional states from the subtypes.

The regional resilience shown by microglia in the middle temporal gyrus shows that not all cortical regions experience extreme reprogramming during disease progression. It underscores the importance of analyzing both vulnerable and resilient regions of the brain to combat the growth of disease.

The analysis of RNA-sequenced data for microglia and neurons in Alzheimer’s and Parkinson’s diseases exemplified processes such as immune activation, pathway disruptions, transcriptional shifts and more that are associated with neural degeneration. Microglial cells in the Middle Temporal Gyrus showed enrichment patterns correlating with neuroinflammation and cellular stress, while also displaying resilience and structural stability in the event of Alzheimer's Disease. These changes show the specificity of cell-type transcriptomes in different regions of the brain under stress and degeneration. In the future, more research can be done that dives deep into how other cell types are affected in different regions of the brain that may not have been explored as much before, which can help guide therapeutic strategies in resolving neural dysfunction.

MATERIALS AND METHODS

Gene Set Enrichment Analysis and Pathway Filtering

The differential results for the expressions of Alzheimer's and Parkinson's Diseases were used for both microglia and neurons. These samples were used for GSEA (Gene Set Enrichment Analysis), using Gene Ontology (GO) Biological Processes. The genes were ranked by differential expression statistics, and the GO process was implemented through South Dakota State's ShinyGo bioinformatics enrichment tool. An FDR cutoff of 0.05 was implemented to ensure that only significant enriched genes were used. For both microglia and neurons, separate analyses were conducted for both cell types, noting down the pathways based on the upregulated and downregulated genes. Based on the graphs, the number of genes for each pathway was based on the size of the black point on the legend, and the statistical significance in comparison to other pathways was determined by the color gradient.

Differential Expression and Threshold Visualization

For constructing a volcano plot for both microglia and neurons in Alzheimer's disease, differential expression analysis was performed to compare the gene expressions and activations between cell types in different disease groups. An effect size threshold of 1 and a p-value of 0.05 was used. These thresholds were implemented to ensure only significant genes were used in the analysis. Having the thresholds the same for both cell types was a way to ensure that there would be no bias in the results of significant genes and results. The volcano plots were constructed thereafter, to show the correlation of the log fold change and the adjusted p-values. Using online software and tools such as Plotly, the violin plots were constructed.

Acquisition of MTG scRNA-seq Data Set

Our team downloaded data from the Seattle-AD repository regarding Microglia in the Middle Temporal Gyrus through CellxGene, The transcriptomic data contained 40,000 cells, and 36,412 expressed genes, and a 5% cutoff was implemented to exclude cells with excessive mitochondrial content. Following the *Single-Cell Best Practices* pipeline, all analyses were formed with Scanpy, and Python, V3.11. Our team initially performed Quality Control, to remove both duplicated Unique Molecular identifiers, as well as low quality cells. The metrics that we used for our code were n_genes_by_counts, total_counts, as well as the fraction of Mitochondrial UMI's.

Pre-Processing, Clustering, Annotation, Data Integration Implementation

Following the Single Cell Practices preamble, clustering, annotations, dimensionality reduction were performed respectively. After the HVG's were identified with the Seurat flavor, and the Dimensionality Reduction was implemented to display trends, Leiden Clustering was applied. Both methods were visualized through UMAP, and a resolution of approximately 0.5 was used. The clustering was for identifying different subgroups of the Microglial cell types in Alzheimer's, dependent on connectivity between each cell in the Middle Temporal Gyrus.

Marker-Based Annotation and Donor Integration

Using established microglial marker genes, cells were annotated to confirm the identities were microglia based before the donor integration. Following annotation, the donor level metadata was incorporated to find batch and donor effects across the microglial dataset. Integrated donor information was plotted on the UMAP, and overlap among the donor identities confirmed a successful integration.

Comparison of Alzheimer's and Control Transcriptomes

Using Donor level metadata of microglia in the MTG, the dataset was integrated, derived from the SEA-AD study for Alzheimer's. Alzheimer's and Control were plotted and visualized separately in order to compare and contrast any glaring differences between the donor-level data sets. Following the SC-Best-Practices workflow, the same batch correction was used from the previous diagram, using the same Leiden clustering.

ACKNOWLEDGMENTS (Optional)

Authorship is considered the highest form of acknowledgement in scientific writing. This means that anyone in your author list cannot appear in your acknowledgements. Include anyone or anyplace that helped with your research but didn't meet the requirements for authorship.

If you received any funding for your work this is where you want to acknowledge it!

REFERENCES

Agrawal, Ishan, and Sushmita Jha. "Mitochondrial Dysfunction and Alzheimer's Disease: Role of Microglia." Frontiers in Aging Neuroscience, vol. 12, Aug. 2020, p. 252. PubMed Central, https://doi.org/10.3389/fnagi.2020.00252.

American Brain Foundation. Disease Connections: Alzheimer's and Parkinson's. 6 Feb. 2024, https://www.americanbrainfoundation.org/disease-connections-alzheimers-and-parkinsons/.

Bachstetter, Adam D., et al. "Disease-Related Microglia Heterogeneity in the Hippocampus of Alzheimer's Disease, Dementia with Lewy Bodies, and Hippocampal Sclerosis of Aging." Acta Neuropathologica Communications, vol. 3, no. 1, May 2015, p. 32. Springer Link, https://doi.org/10.1186/s40478-015-0209-z.

Gabitto, Mariano I., et al. "Integrated Multimodal Cell Atlas of Alzheimer's Disease." Nature Neuroscience, vol. 27, no. 12, Dec. 2024, pp. 2366–83. www.nature.com, https://doi.org/10.1038/s41593-024-01774-5.

Gjoneska, Elizabeta, et al. "Conserved Epigenomic Signals in Mice and Humans Reveal Immune Basis of Alzheimer's Disease." Nature, vol. 518, no. 7539, Feb. 2015, pp. 365–69. PubMed, https://doi.org/10.1038/nature14252.

Hammond, Timothy R., et al. "Single-Cell RNA Sequencing of Microglia throughout the Mouse Lifespan and in the Injured Brain Reveals Complex Cell-State Changes." Immunity, vol. 50, no. 1, Jan. 2019, pp. 253-271.e6. DOI.org (Crossref), https://doi.org/10.1016/j.immuni.2018.11.004.

Ilicic, Tomislav, et al. "Classification of Low Quality Cells from Single-Cell RNA-Seq Data." Genome Biology, vol. 17, no. 1, Feb. 2016, p. 29. Springer Link, https://doi.org/10.1186/s13059-016-0888-1.

Kim, Kyoung Mi, et al. “Mitochondrial RNA in Alzheimer’s Disease Circulating Extracellular Vesicles.” Frontiers in Cell and Developmental Biology, vol. 8, Nov. 2020. Frontiers, https://doi.org/10.3389/fcell.2020.581882.

Lovinger, David M. “Neurotransmitter Roles in Synaptic Modulation, Plasticity and Learning in the Dorsal Striatum.” Neuropharmacology, vol. 58, no. 7, June 2010, pp. 951–61. PubMed Central, https://doi.org/10.1016/j.neuropharm.2010.01.008.

Mayo Clinic. Alzheimer’s Disease - Symptoms and Causes. https://www.mayoclinic.org/diseases-conditions/alzheimers-disease/symptoms-causes/syc-20350447. Accessed 12 Jan. 2026.

National Institute of Neurological Disorders and Stroke. “Parkinson’s Disease | National Institute of Neurological Disorders and Stroke.” National Institute of Neurological Disorders and Stroke, https://www.ninds.nih.gov/health-information/disorders/parkinsons-disease. Accessed 12 Jan. 2026.

Olah, Marta, et al. “Single Cell RNA Sequencing of Human Microglia Uncovers a Subset Associated with Alzheimer’s Disease.” Nature Communications, vol. 11, no. 1, Nov. 2020, p. 6129. www.nature.com, https://doi.org/10.1038/s41467-020-19737-2.

Pfeil, Julia, et al. “Unique Regional Patterns of Amyloid Burden Predict Progression to Prodromal and Clinical Stages of Alzheimer’s Disease.” Neurobiology of Aging, vol. 106, Oct. 2021, p. 119. pmc.ncbi.nlm.nih.gov, https://doi.org/10.1016/j.neurobiolaging.2021.06.014.

Prater, Katherine E., et al. “Human Microglia Show Unique Transcriptional Changes in Alzheimer’s Disease.” Nature Aging, vol. 3, no. 7, July 2023, pp. 894–907. www.nature.com, https://doi.org/10.1038/s43587-023-00424-y.

Salta, Evgenia, et al. “Adult Hippocampal Neurogenesis in Alzheimer’s Disease: A Roadmap to Clinical Relevance.” Cell Stem Cell, vol. 30, no. 2, Feb. 2023, pp. 120–36. PubMed Central, https://doi.org/10.1016/j.stem.2023.01.002.

Science Direct. Middle Temporal Gyrus - an Overview | ScienceDirect Topics. https://www.sciencedirect.com/topics/neuroscience/middle-temporal-gyrus. Accessed 12 Jan. 2026.

Stratoulias, Vassilis, et al. “Microglial Subtypes: Diversity within the Microglial Community.” The EMBO Journal, vol. 38, no. 17, Sept. 2019, p. e101997. PubMed Central, https://doi.org/10.15252/embj.2019101997.

Sun, Na, et al. “Human Microglial State Dynamics in Alzheimer’s Disease Progression.” Cell, vol. 186, no. 20, Sept. 2023, pp. 4386-4403.e29. DOI.org (Crossref), https://doi.org/10.1016/j.cell.2023.08.037.

Figures and Figure Captions

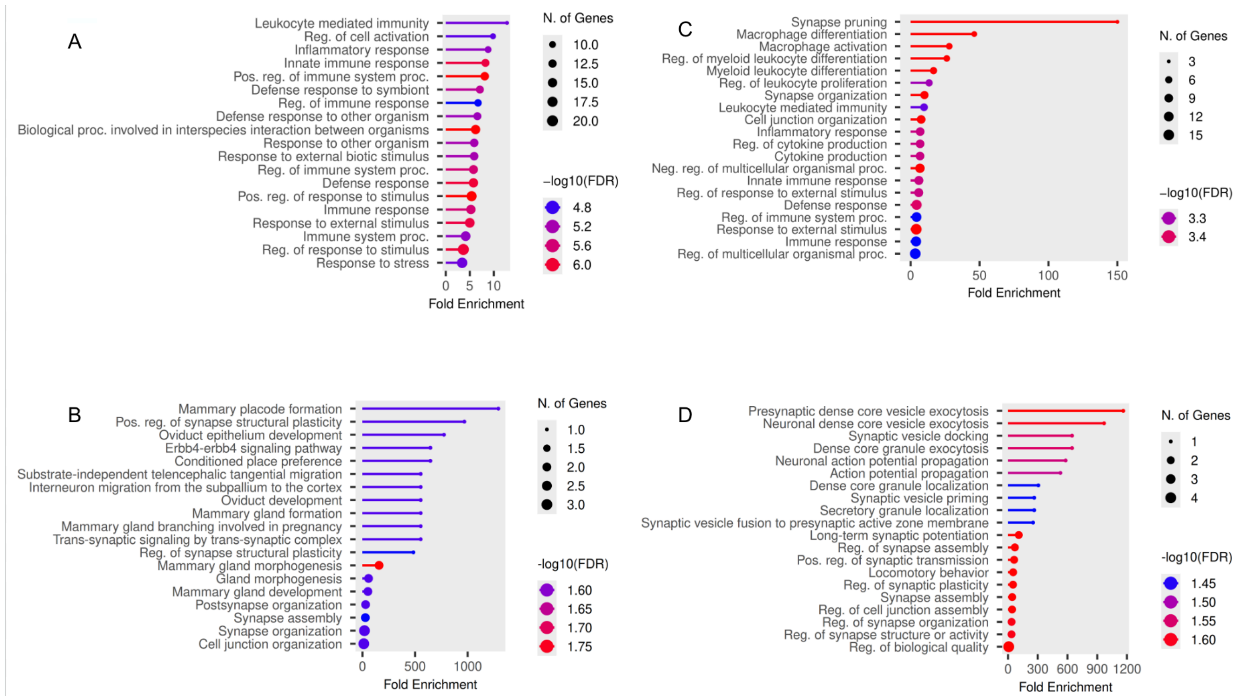

Figure 1. Gene Set Enrichment Analysis of different genes (GO Biological Process) of neurons and microglia in Alzheimer's & Parkinson's Disease. (**A**) Upregulated Microglia in both Alzheimer's & Parkinson's Diseases show enrichment in immune related pathways **(B)** Downregulated Microglia in both Alzheimer's & Parkinson's Diseases show enrichment in differentiation and the regulation of immune pathways **(C)** Upregulated neurons in both Alzheimer's & Parkinson's Diseases show enrichment in synaptic processes **(D)** Downregulated neurons in both Alzheimer's & Parkinson's Diseases show enrichment exocytosis processes.

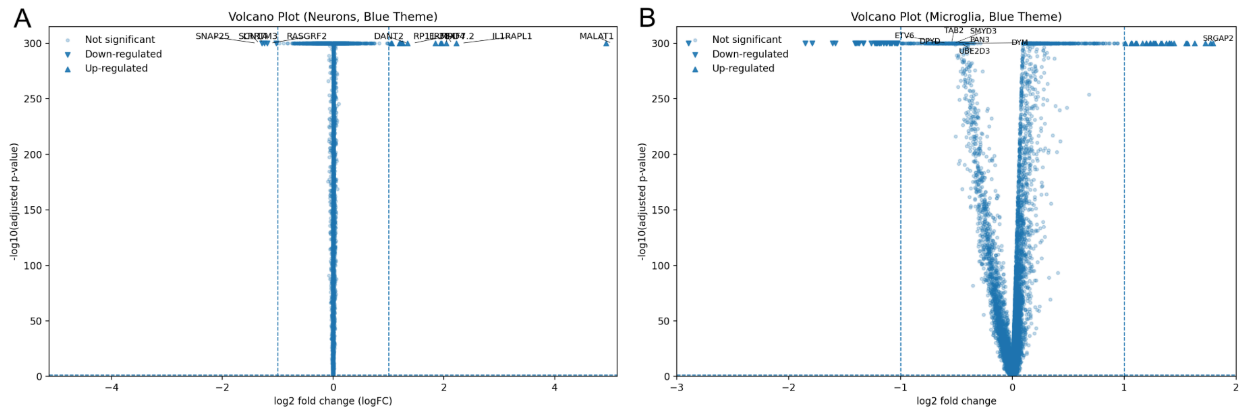

Figure 2. Differential gene expression analysis of neurons and microglia in Alzheimer's and Parkinson's disease. (A) Volcano plot showing $\log_2$ fold change versus $-\log_{10}$ (adjusted p-value) for neurons. **(B)** Volcano plot showing $\log_2$ fold change versus $-\log_{10}$ (adjusted p-value) for microglia. Each dot represents an individual gene tested for differential expression between Alzheimer's and Parkinson's disease samples. Highly significant genes are labeled near the top of each plot. In the study, there was an effect size of 1, and a p value of 0.05 used.

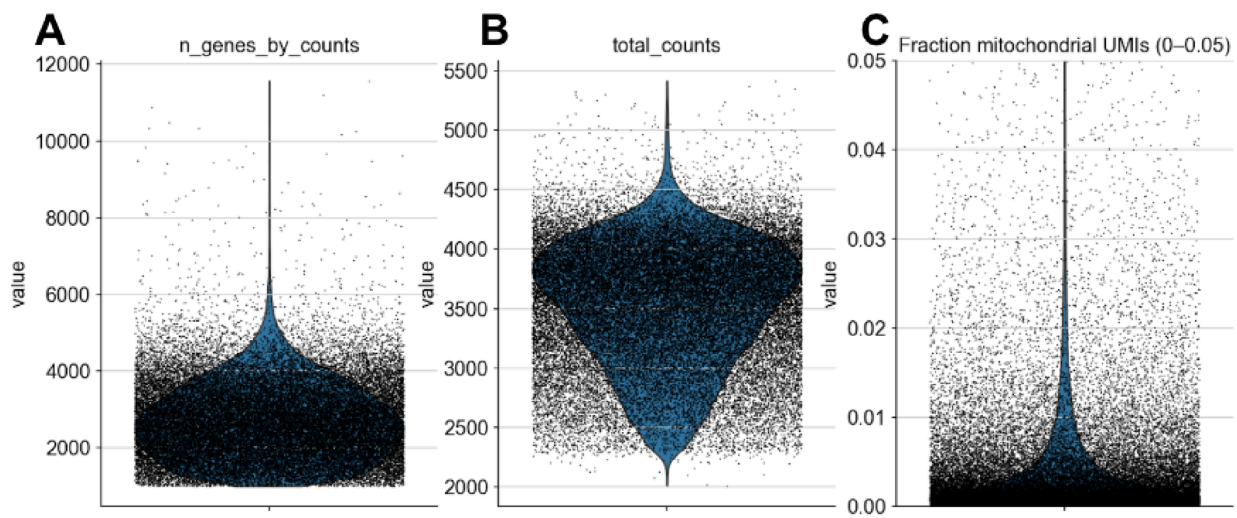

Figure 3. Summary data and quality control of scRNA-seq data from microglia in the middle temporal gyrus. (**A**) Number of genes detected per cell. (**B**) Total amount of unique molecular identifiers per cell. (**C**) Fraction of unique molecular identifiers that mapped to mitochondrial genes per cell. Each dot represents a microglial cell. The blue shaded violin plot indicates distribution for each metric.

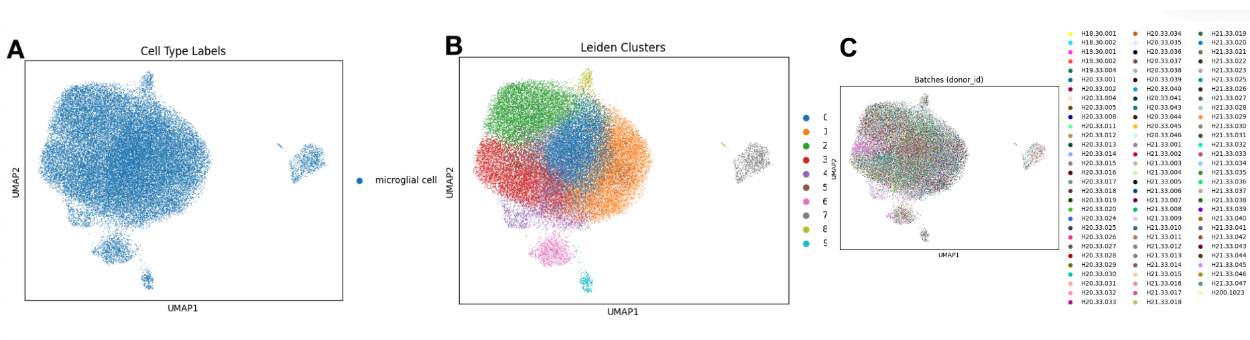

Figure 4. UMAP clustering and annotated labeling of microglial cells from MTG (Middle Temporal Gyrus) (A) UMAP plot shows each microglial cell, associated with the Donor ID, as well as batch distributions. **(B)** UMAP labeling, determining each cell used as a Microglial-type cell from each donor, as well as confirming batch distributions. **(C)** UMAP projection of batch distribution associated with each donor, revealing 10 subtypes of Microglia.

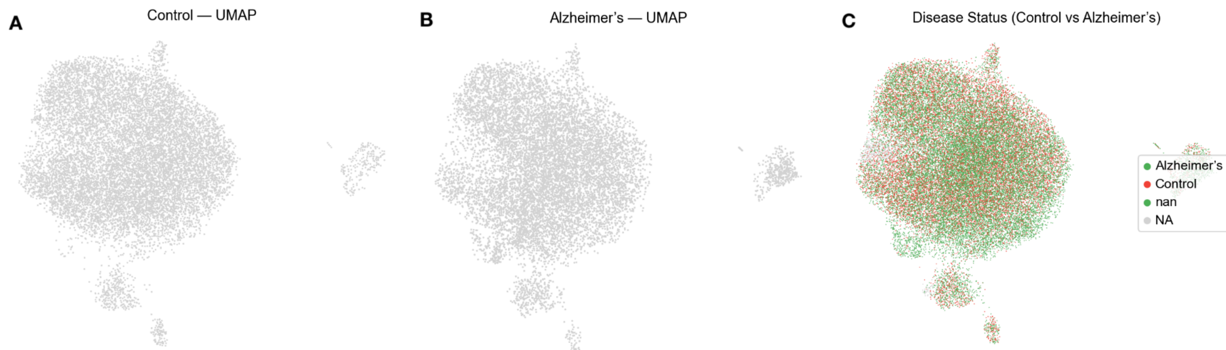

Figure 5. UMAP Visualization of disease status & Control Vs. Disease. **(A)** UMAP visualization of microglial cells derived from control donors. **(B)** UMAP visualization of microglial cells derived from Alzheimer's Disease donors. **(C)** Combined UMAP overlap representing control and Alzheimer's microglia, annotated by disease status.